REJIT B ANTONY

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Education

Vellore Institute of Technology Chennai, India

B.Tech – Computer Science Engineering

July 2023 – Present

CGPA: **9.12**

Navdeep Public School Kollam, Kerala

High School – CBSE

May 2021 – May 2023

Graduated with **94.2%** in Grade 12

Skills

Languages

Python • Java • JavaScript • LaTeX

Frameworks & AI

Gemini API • PyTorch • TensorFlow • Keras • React • Astro • Node.js • Express.js • Scikit-learn • Pandas • NumPy

Databases

MongoDB • SQL • SQLite3

Tools

Claude Code • OpenCode • Gemini CLI / Agents • Git • VS Code • JupyterLab • Adobe Photoshop • Unity • Godot

Operating Systems

Linux (Arch, WSL, CLI) • Windows

Certifications

- **1st Place – FOSSIT** (LUGVITC, April 2024)
- **Volunteer Recognition Award** – Raptee.HV, CodeVolt (Mar 2025)
- **Full Stack MERN** Certification (Delta Course) – ApnaCollege
- **Agile & Design Thinking** – IBM
- **DevOps Fundamentals** – IBM

Projects

PICSHARE: AI-Powered Cloud Photo Gallery

GitHub

Open-Source Contribution

- Cut **lightbox image load** from ∼**2.4 s to** <**24 ms** (median) by serving pre-generated 2K WebP previews from local disk instead of streaming originals from Google Drive, with files ∼**99% smaller** (**30 MB** → ∼**270 KB**).
- Built a **Drive-backed image store** (Drive as source of truth; 2K WebP previews, thumbnails, and face embeddings on the server), cutting server disk from ∼**300 GB** of originals to ∼**3.3 GB** per 10k photos.
- Built a TypeScript **Canvas compressor** that shrinks multi-MB selfies to ∼**300 KB** before upload, accelerating uploads and avoiding proxy payload errors without hurting **InsightFace** matching.

Reinforcement Learning – BipedalWalker (TD3 & TD3-FORK)

GitHub

PyTorch • Gymnasium • CUDA

- Implemented TD3 from scratch in PyTorch (replay buffer, twin critics, actor, Polyak target updates) with no RL library dependencies; trained on BipedalWalker-v3 to avg reward **+280/300** in **1M steps** on personal GPU.
- Built a live **6-panel** Matplotlib training dashboard (reward curves, Q-value drift, action distributions, TD error) and a threaded `pygame` render window with HUD overlay displaying training in real time without blocking the training loop.
- Extended the agent with TD3-FORK (forward dynamics model predicting next state from current state-action) to address long-horizon credit assignment failure on BipedalWalkerHardcore-v3; reward improved from **-40 to +100+** within **900k steps**.

Placement Tracker Platform (Full-Stack Features & Data Pipelines)

GitHub

TypeScript • Svelte • SQLite3

- Architected data import pipelines processing **15,000+** records via scripts, ensuring robust **data integrity**.
- Executed a structural database migration from legacy student IDs to a unified **NeoID** architecture, implementing efficient SQL joins and pagination to slash query latency from **2000ms to 50ms**.

Image Denoising Autoencoder

GitHub

TensorFlow • Keras • Python

- Built a hybrid Inception-ResNet denoising autoencoder in TensorFlow/Keras trained on DIV2K, using Inception blocks **(1×1, 3×3, 5×5)** and stacked ResNet residual blocks across a **3-stage** encoder.
- Implemented a U-Net style decoder with skip connections and mixed-precision training (`float16`) to preserve spatial detail while halving activation VRAM consumption.
- Engineered a memory-efficient pipeline using a custom Keras `Sequence` generator for on-the-fly noise augmentation across **5000 patches** from **800 source images**.

Extracurricular Activities

The Linux Club

VIT Chennai

Media Lead June 2024 – April 2026

- Designed and contributed to event frontends. Managed onground media team. Maintained 100+GB event footage and photos.

The Robotics Club

VIT Chennai

Media & Design Lead April 2025 – April 2026

- Designed visual branding and led live event photo/video coverage.

CodeVolt '25 Hackathon

VIT Chennai

Volunteer March 2025

- Supported event coordination, ensuring a seamless experience for participants and judges and assisted teams with technical queries and troubleshooting during the hackathon.